

Assignment 2 – Python programming

1) Define a procedure histogram() that takes a list of integers and prints a histogram to the screen. For example, histogram([4, 9, 7]) should print the following:

```
****
*****
*****
```

2) Write a function which accepts a string and prints out
i) the number of alphanumeric characters present in the string
ii) the number of words present in the string

3) Write a program that computes the net amount of a bank account based a transaction log from console input. The transaction log format is shown as following:

```
D 100
W 200
```

D means deposit while W means withdrawal.

Suppose the following input is supplied to the program:

```
D 300
D 300
W 200
D 100
```

Then, the output should be: 500

4) Repeat problem 3 with the transactions coming from a file that you need to read and process the data.

5) Write a function which takes in a number as argument and checks whether it is a prime or not.

6) Define a function which can generate a list where the values are square of numbers between 1 and 10 (both included). Then the function needs to print all values except the first 5 elements in the list.

7) Please write a program to generate all sentences where subject is in ["I", "You"] and verb is in ["Play", "Love"] and the object is in ["Hockey", "Football"] and print them to a file.

Sentence format is <subject> <verb> <object>